

THE BRAIN BRAKE

Version six, boarded. What each department makes, frame by frame.

Eight scenes, forty eight frames, two minutes. Every frame in the film is on the following pages with three things beside it: what the camera has to shoot, what the animator has to draw, and what Manan has to do and say.

The film separates evidence from explanation and never mentions it. Manan and his hands are photographed, because they are what is true. Everything else is drawn, because it is what is thought. Nobody in the audience is told this and everybody absorbs it in the first twenty seconds.

THE SPLIT

- 48** frames boarded in total
- 12** frames need Manan in front of a camera
- 36** frames are drawn and need nobody on the day
 - 1** shooting day, in Pondicherry, with Venkatesh
 - 4** lighting setups, and two of them share one lighting state
 - 6** lines he speaks to camera, the rest is voice over

So three quarters of the film is made in Zagreb and never needs him. The day itself is short, and almost all of it is one boy against a plain grey wall doing very little.

The shooting day

Manan is filmed against a plain grey background and placed into the drawn world afterwards, so the shoot itself is simple. Lines are shot one sentence at a time and nothing is ever rushed.

A Grey backdrop, mid shot, walking and reacting

Camera at eye level on sticks, locked off. Soft key from camera left at forty five degrees, large source. Fill from the right at half strength. Slightly warm, because every scene in this setup ends with warm light arriving. Frames 1.5, 1.6, 3.2, 3.3, 3.4, 3.6.

B Seated eyeline, reacting to Coach Brain

The same lighting as A, unchanged. A tennis ball on a stand at seated height, camera left, for the eyeline. Do not relight between A and B, only move the mark. Frames 4.2, 4.6.

C White void, still, to lens

Flat even light from both sides, no modelling, no shadow on the backdrop. The brightest setup of the day and the calmest. Frames 8.1, 8.3, 8.5, 8.6.

D Jump, high frame rate

Same as C, wider, room above his head. Not boarded, five minutes, worth having. Shoot it last when he is loose.

THE ONE THING THE WHOLE LOOK DEPENDS ON

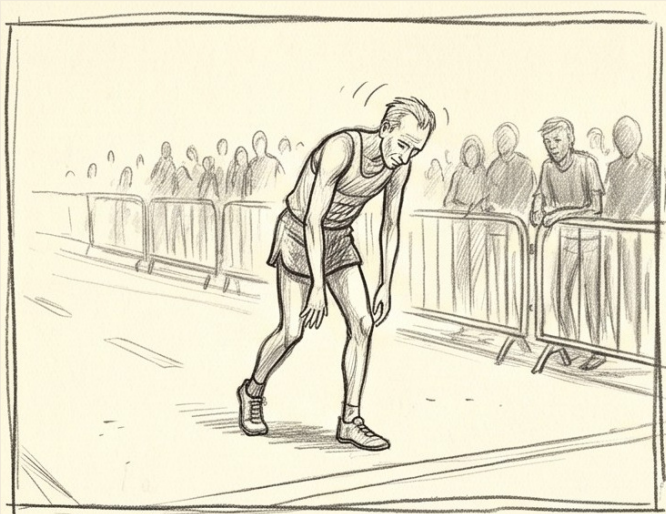
In setups A and B, Manan is lit by a single dominant key from a consistent side, hard enough that his shadow has a shape. That shadow is drawn onto the paper afterwards, and it is what makes him stand inside the drawing instead of sitting on top of it. If the light on the day is flat, every frame in the film loses this and it cannot honestly be added back later.

FOR MANAN, BEFORE THE DAY

Six lines to camera, none longer than two sentences, and the voice over on top of them. He should know them well enough to forget them, because every one of them is written to sound like somebody working something out rather than somebody explaining it. The best takes on this film will be the ones where he is thinking, not the ones where he is right.

Two moments carry more than the rest. Frame 3.6, where he turns to us in the hall of tanks, and frame 8.1, where he stands still with his eyes closed and does nothing at all. The first is the one the judges read him on. The second is the hardest thing on the call sheet.

THE MYSTERY



1.1 DRAWN, nobody needed on the day



1.2 DRAWN, nobody needed on the day



1.3 DRAWN, nobody needed on the day

The runner at the end of himself

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Full drawn wide. He is finished: head rolling, legs gone, the crowd thin and quiet behind the barriers, the road running away to the horizon. Shot 6.4 rhymes this frame exactly, same composition and opposite feeling, so the camera position chosen here is locked for the rest of the film.

MANAN

Voice over only. "For a hundred years we thought the answer was in the muscles."

His face

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Drawn close up, sweat and breath. He is lean and about thirty five. He is at the end of a race, not at the end of a life, and the whole first scene fails if he reads as an old man.

MANAN

Voice over continues.

He explodes into a sprint

CAMERA · VENKATESH

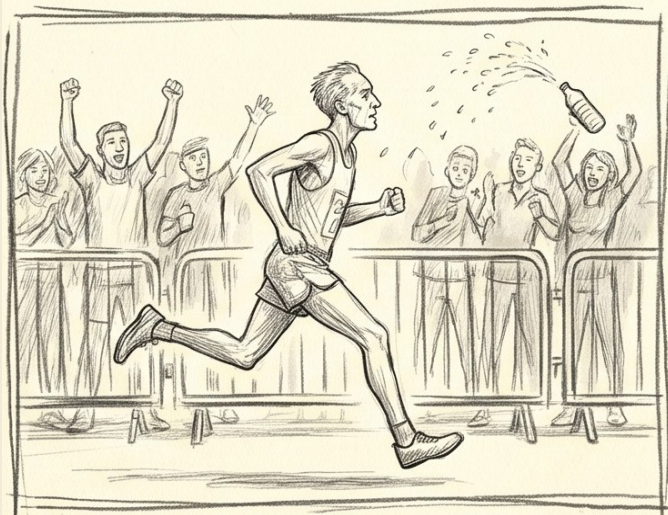
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

The impossible thing, and it happens without warning. Diagonal, violent, left to right, horizon low so he sits high in frame. Speed lines, grit lifting, the crowd coming up off the barriers.

MANAN

Nothing. Let the picture do it.



1.4 DRAWN, nobody needed on the day

The sprint carries

CAMERA · VENKATESH

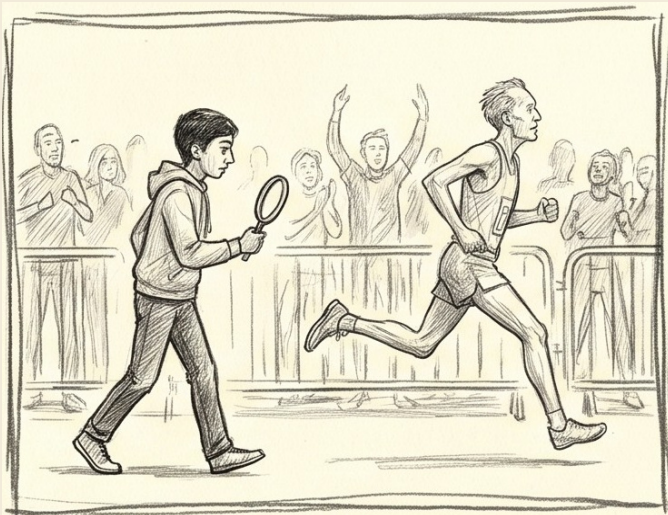
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Held on the sprint, a bottle thrown, the crowd on its feet. Two or three seconds of pure disbelief before the film stops dead.

MANAN

Nothing.



1.5 LIVE ACTION + DRAWN

Everything freezes and Manan walks in

CAMERA · VENKATESH

SETUP A. Grey backdrop, mid shot, camera at eye level on sticks, locked off. He walks slowly across frame looking left and right at nothing at all. Shoot thirty seconds of pure observing with no lines. One dominant key from camera left, hard enough that his shadow has a shape, because that shadow gets drawn onto the road afterwards and it is the single thing the whole look depends on.

ANIMATION · KRISTI JAN

The world stops: crowd, runner, thrown bottle, all held. Manan is cut in and his shadow is drawn onto the tarmac, crossing the pencil lines so they stay visible through it. That shadow is what makes him stand in the drawing instead of on top of it.

MANAN

Walking through a stopped world with the magnifying glass raised, looking at things. Curious, taking his time. No line here. He is not performing for anyone, he is just looking.



1.6 LIVE ACTION + DRAWN

He turns to us and asks the question

CAMERA · VENKATESH

SETUP A. Close up, same light, to lens. Six takes minimum and use the most unguarded.

ANIMATION · KRISTI JAN

The frozen crowd stays behind him. Title over.

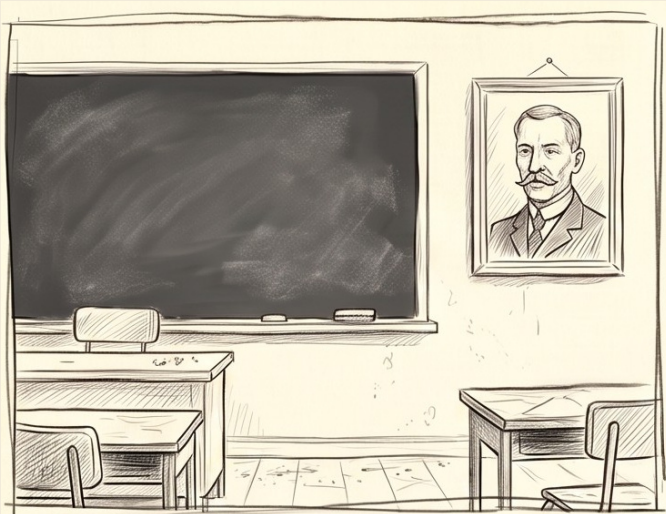
MANAN

He lowers the glass and turns to camera.

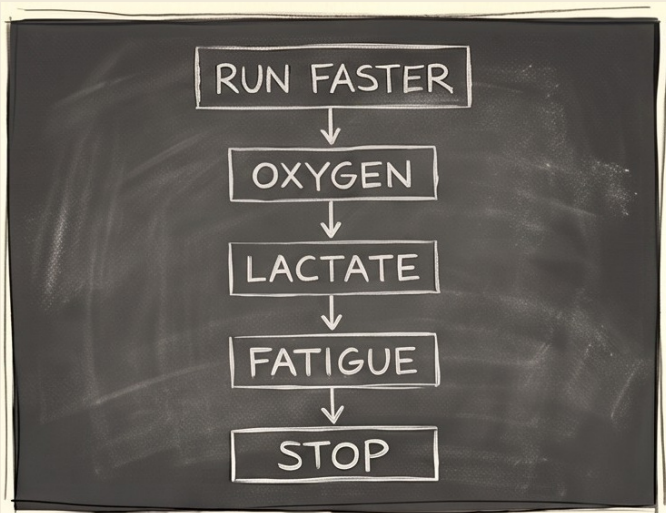
"Hold on. He had nothing left. So where did THAT come from?"

Curious and amused, never presenting. He has just noticed something, he is not announcing it.

THE OLD THEORY



2.1 DRAWN, nobody needed on the day



2.2 DRAWN, nobody needed on the day



2.3 DRAWN, nobody needed on the day

1923, and the man who answered it

CAMERA • VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION • KRISTI JAN

A lecture room of the period, empty desks, blackboard, A. V. Hill's portrait on the wall. Warm and respectful. There is no villain in this film and this scene is where that is decided.

MANAN

Voice over. "A. V. Hill, Nobel prize, 1923."

The chain that held for a century

CAMERA • VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION • KRISTI JAN

The chain writes itself on the board one box at a time: RUN FASTER, OXYGEN, LACTATE, FATIGUE, STOP. Chalk the year 1923 at the top of the board. This panel will be screenshotted on its own, and without the year on it the board states a hundred year old theory as current fact.

MANAN

Voice over. "Run hard, run out of oxygen, and the muscle stops."

Inside the leg, a working factory

CAMERA • VENKATESH

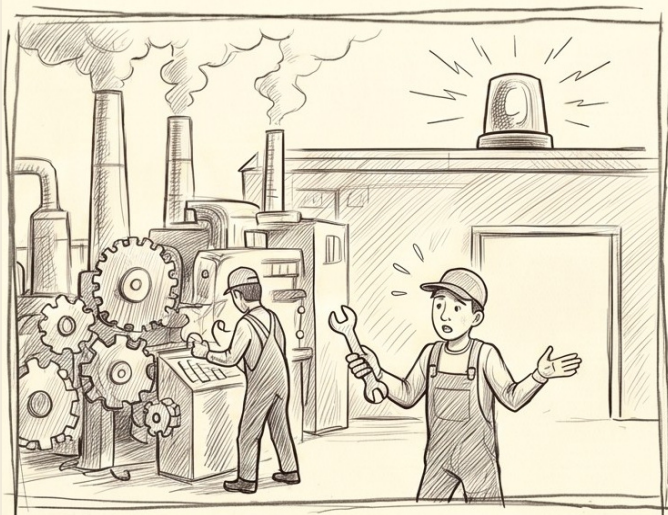
Nothing on the day. This frame is drawn.

ANIMATION • KRISTI JAN

The muscle as an industrial plant, gears turning, belts running, crates arriving, a delivery of glucose at the loading bay. Everything healthy and in rhythm. This is Manan's own image from his first draft and it earns its place by being the clearest picture of the old theory anyone has drawn.

MANAN

Voice over.



2.4 DRAWN, nobody needed on the day

The supply stops

CAMERA · VENKATESH

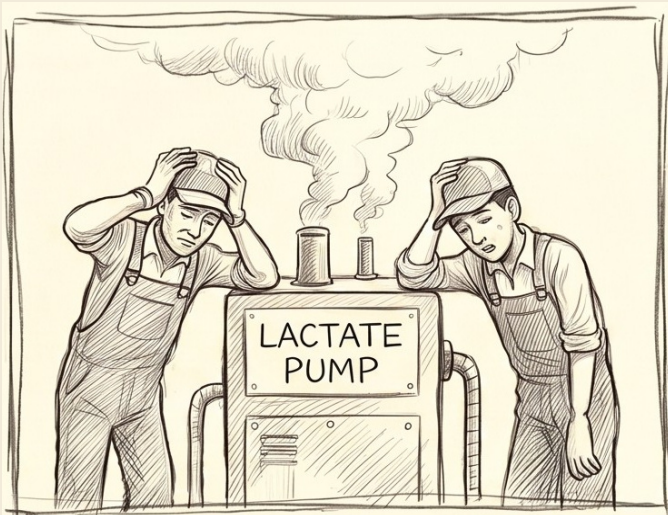
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

The alarm turns, a worker straightens up and shrugs, palms open. The supply is failing, the machine is not breaking. Resigned, not frightened. This is a shift ending, not a disaster.

MANAN

Worker's line, one voice, dry. "That's it. We're done."



2.5 DRAWN, nobody needed on the day

The lactate pump

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Two workers at the pump with their hands on their heads, steam going up. The old theory's villain, lactate, drawn exactly as the theory imagined it.

MANAN

Voice over.



2.6 DRAWN, nobody needed on the day

Case closed, and the crack

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

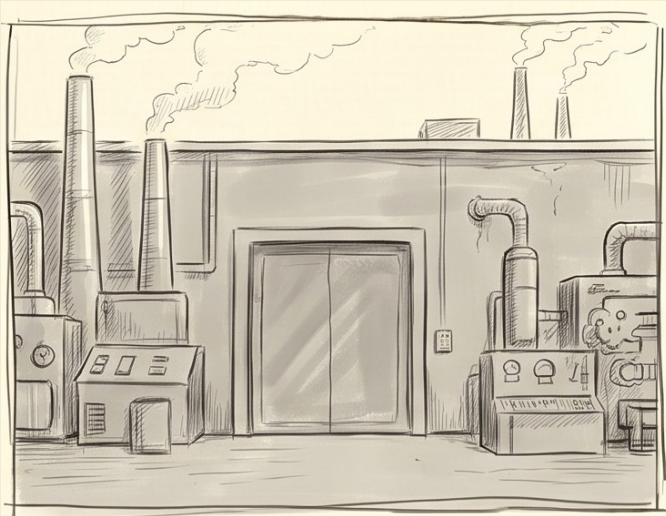
ANIMATION · KRISTI JAN

The stamp comes down over the whole factory with real weight. Then a fine crack runs across the face of it and the sprint replays inside the crack. The crack is the scene. Everything before it exists to make this land.

MANAN

Voice over. "Case closed. For a hundred years." Then, over the crack: "Except it doesn't explain this."

THE FULL TANK



3.1 DRAWN, nobody needed on the day

The back of the factory

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIYAN

Wide of the plant from outside, and one door in it that nobody has opened. Quiet after the noise of scene two.

MANAN

Voice over.



3.2 LIVE ACTION + DRAWN

His hand on the door

CAMERA · VENKATESH

SETUP A. He stands with his back to camera and puts his palm flat on a mark at chest height, head tilted, listening. There is nothing really there. Hold it, then let him lean into it.

ANIMATION · KRISTIYAN

The riveted door is drawn around his hand, with a keyhole beside it and light behind the keyhole. His shadow falls on the door.

MANAN

Back to camera, palm flat, listening. He is curious rather than cautious.

Voice over. "At the back of the factory there is a door nobody opens."



3.3 LIVE ACTION + DRAWN

The door swings

CAMERA · VENKATESH

SETUP A. Silhouette from behind, then turning into the light. Shoot it both ways.

ANIMATION · KRISTIYAN

The door opens and warm light floods out over him. This is the first warm light in the film and it arrives as a discovery, not as a threat.

MANAN

He pushes and steps into the light. No line.



3.4 LIVE ACTION + DRAWN

The hall of tanks

CAMERA · VENKATESH

SETUP A. Back of his head and shoulder in the near foreground, held still.

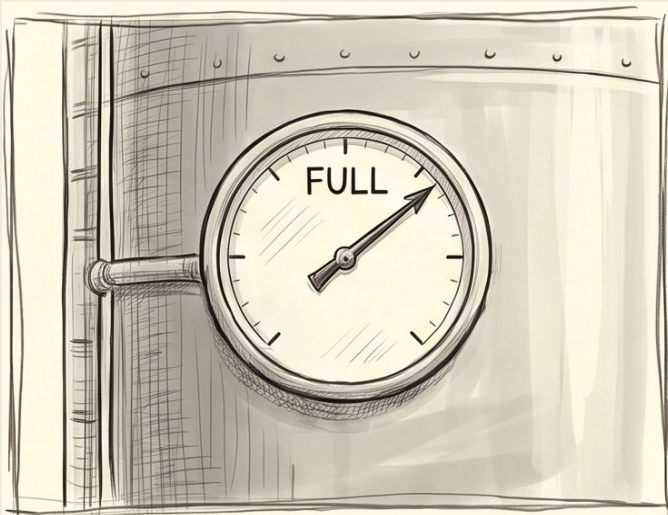
ANIMATION · KRISTI JAN

Row upon row of storage tanks receding into the distance. Nine tanks visible, the nearest one stencilled 9. The reveal, and the first big lift in the film. Discovery, not warning.

MANAN

Looking. Nothing else.

Voice over: when scientists tested athletes at the exact moment they gave up, the muscles could still produce far more power than the athlete had just produced.



3.5 DRAWN, nobody needed on the day

Every gauge full

CAMERA · VENKATESH

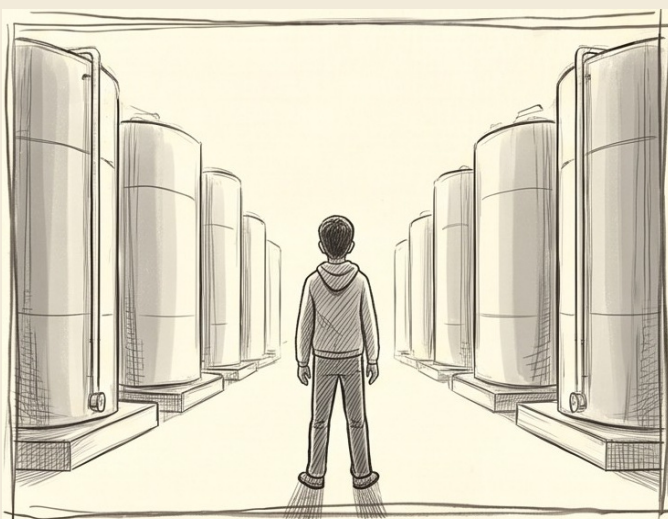
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

One gauge in close up, the needle hard over on FULL. This instrument governs the rest of the film, so establish it clearly here.

MANAN

Nothing.



3.6 LIVE ACTION + DRAWN

He turns to us

CAMERA · VENKATESH

SETUP A. Close up, to lens, lit by the warm light coming from the tanks. This is the most important take of the shooting day. Shoot it many times and use the most unguarded one.

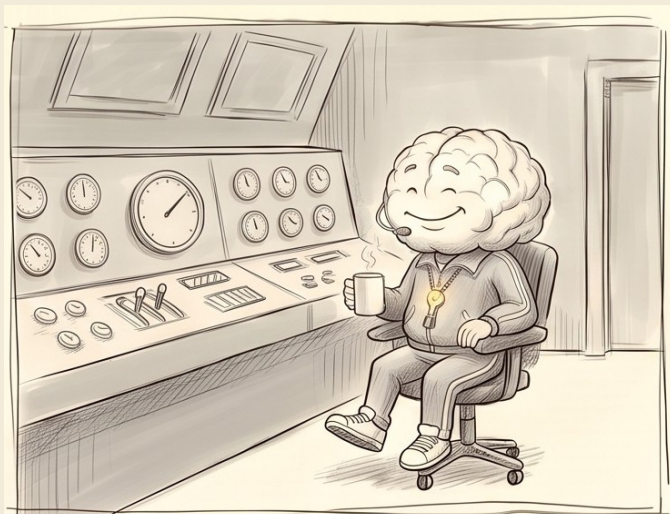
ANIMATION · KRISTI JAN

The hall behind him. He must be turned to the lens in this frame: as boarded, scene three never shows his face once, and this is the scene where the audience decides whether the boy understands what he is saying.

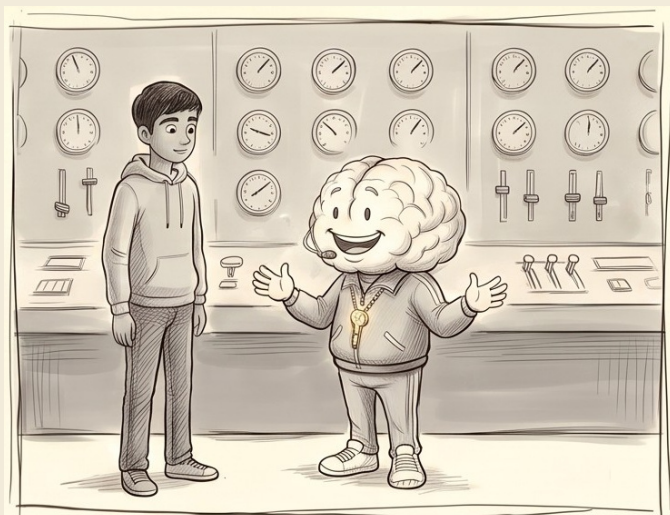
MANAN

Delighted rather than shocked. He has found something, and it is better than what he expected to find. "It was never empty. Somebody closed that door."

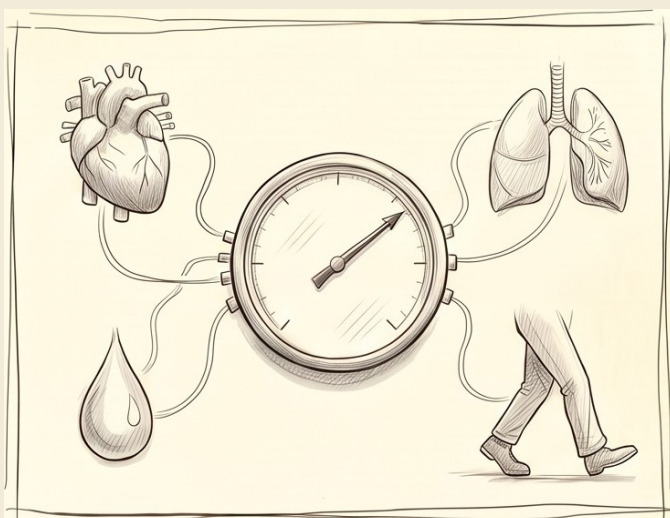
THE GATEKEEPER



4.1 DRAWN, nobody needed on the day



4.2 LIVE ACTION + DRAWN



4.3 DRAWN, nobody needed on the day

Coach Brain, found

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

A warm control room, monitors and dials on every wall. Coach Brain in a tracksuit in a swivel chair with a mug and a small gold key on a chain at his chest. He is not caught out. He is delighted to be found. Establish the key clearly here, it is the object the whole film turns on.

MANAN

Voice over hands over.

COACH BRAIN: "You found me. Took you long enough."

The two of them, level

CAMERA · VENKATESH

SETUP B. Seated eyeline. A tennis ball on a stand at seated height, camera left, for him to look at. Do not relight between setup A and setup B, only move the eyeline mark. Shoot the line, then ten seconds of him simply listening.

ANIMATION · KRISTIJAN

The console wall behind them. The two characters are evenly matched in the frame, the same size and the same weight. Nobody is above anybody here.

MANAN

Facing Coach Brain, level with him.

"You closed that door."

Not an accusation. He is working it out in the moment.

The network

CAMERA · VENKATESH

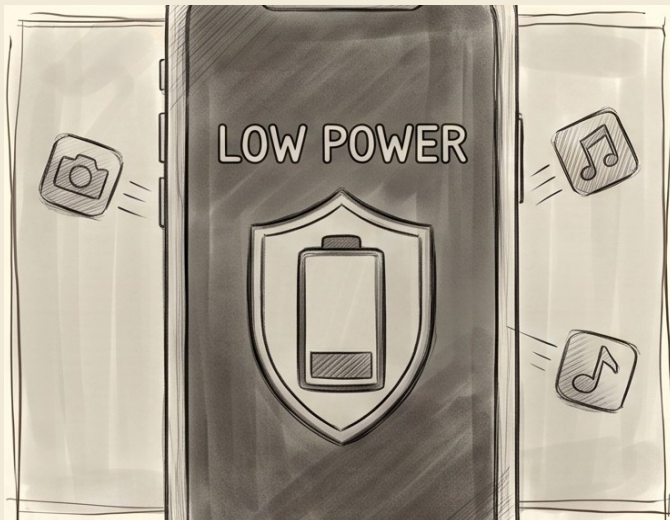
Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

Five readouts converging on one dial: heart, breath, temperature, water, distance. One click as each one connects, five evenly spaced. The dial carries a red arc and twenty seven tick marks.

MANAN

COACH BRAIN: "Heart rate. Breath. Temperature. Water. Distance. I'm asking one question. Can we keep going safely?"



4.4 DRAWN, nobody needed on the day

Low power mode

CAMERA · VENKATESH

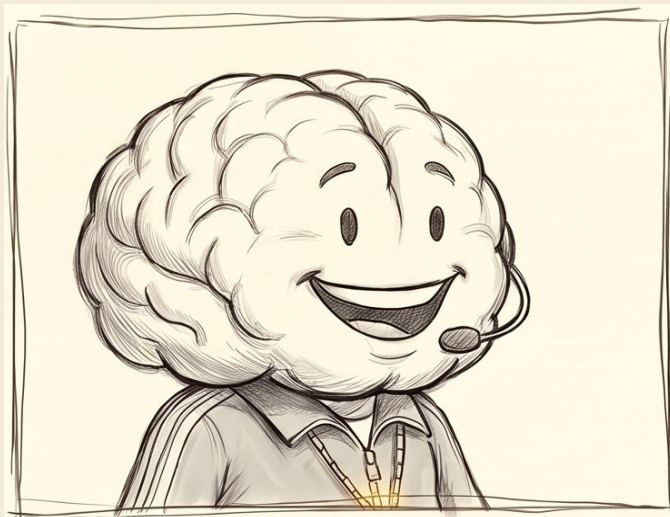
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Two seconds of a phone dimming, apps closing around the edges. A generic phone with no brand anywhere on it. This is Manan's own analogy from his first draft and it is the clearest everyday explanation in either version, so it is kept and compressed.

MANAN

Voice over. "Like a phone at twenty percent. Not broken. Protecting itself."



4.5 DRAWN, nobody needed on the day

He is not sorry

CAMERA · VENKATESH

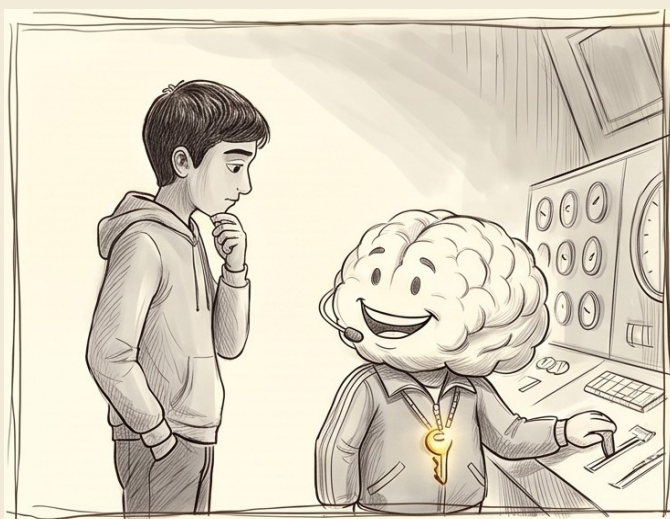
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Coach Brain in close up, wide open and entirely unembarrassed. He is a teacher with a better model of the situation than his student has, and he knows it.

MANAN

COACH BRAIN: "My best judgement. I'm not trying to stop you. I'm trying to get you to the finish line."



4.6 LIVE ACTION + DRAWN

Manan arrives at it

CAMERA · VENKATESH

SETUP B. Same light, same eyeline mark. He should sound like he is discovering the idea, not stating it. Shoot the thinking as well as the line.

ANIMATION · KRISTI JAN

Coach Brain works the console beside him without looking up. A small subtitle low in frame, in and out in two seconds: Central Governor Theory, proposed by Prof. Tim Noakes, 1997, and that scientists still debate how brain and muscle share the work. That honesty is scored by the judges.

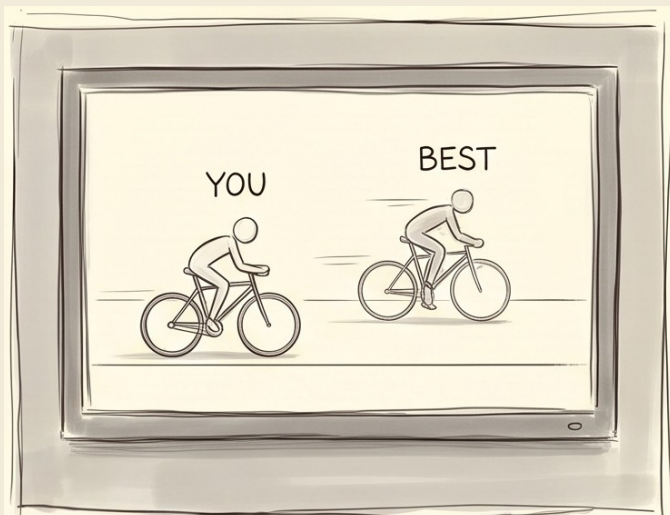
MANAN

In profile, hand at his chin, working it out. "So the limit isn't my body. It's your judgement."

THE TRICK



5.1 DRAWN, nobody needed on the day



5.2 DRAWN, nobody needed on the day



5.3 DRAWN, nobody needed on the day

A cyclist and a ghost

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJA

A clean laboratory, not a clinical one. The cyclist on a stationary bike facing a screen with a translucent version of himself on it.

MANAN

Voice over. "So they let a cyclist race a recording of his own best ride."

YOU and BEST

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJA

The screen, plainly lettered, his ghost ahead of him. These two words and the figure +2% are the only lettering permitted anywhere in this scene.

MANAN

Voice over.

The two percent

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJA

The researcher at the console quietly turning the ghost up. One conspiratorial click and nothing said. +2% on the monitor, lap counter on 27.

MANAN

Voice over. "Except they made the ghost two percent faster. And they didn't tell him."



5.4 DRAWN, nobody needed on the day

He chases

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

The cyclist's face in close up, teeth bared, sweat coming off him. His expression is the shot.

MANAN

Nothing under this. Let it run.



5.5 DRAWN, nobody needed on the day

He draws level

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

The screen again, the two riders overlapping. Cut this scene like a race. The overlap as he comes through is the money frame of the sequence.

MANAN

Nothing.



5.6 DRAWN, nobody needed on the day

He passes his own best

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

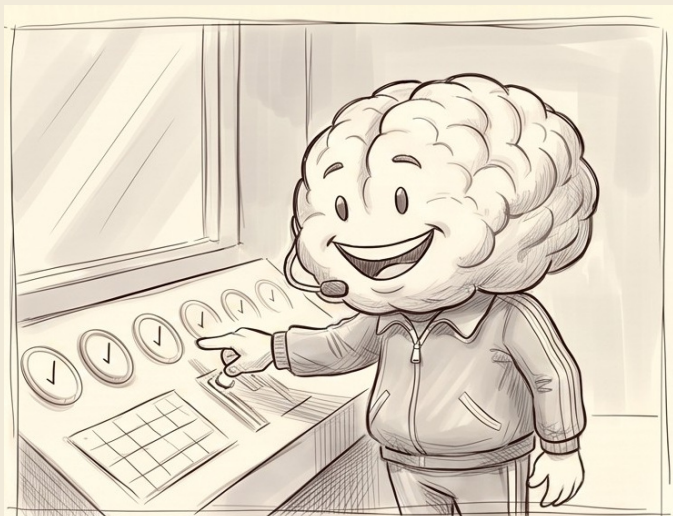
He goes by, spent and astonished, and the ghost dissolves behind him.

MANAN

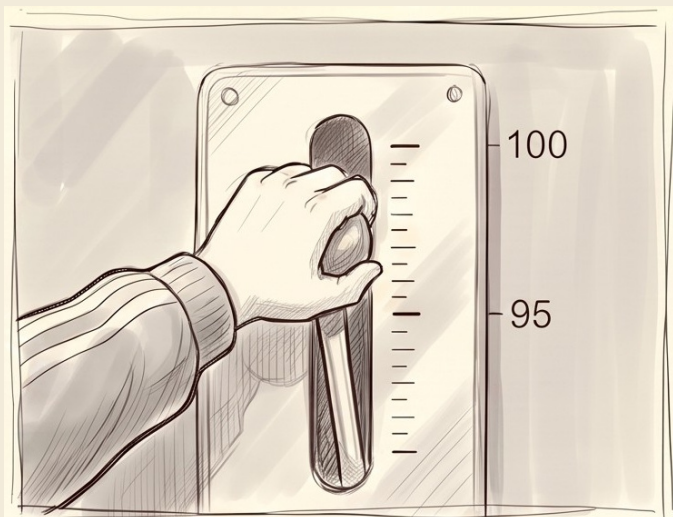
Voice over. "He beat his own maximum. Which means it was never his maximum."

Keep his laugh in the sound if there is one.

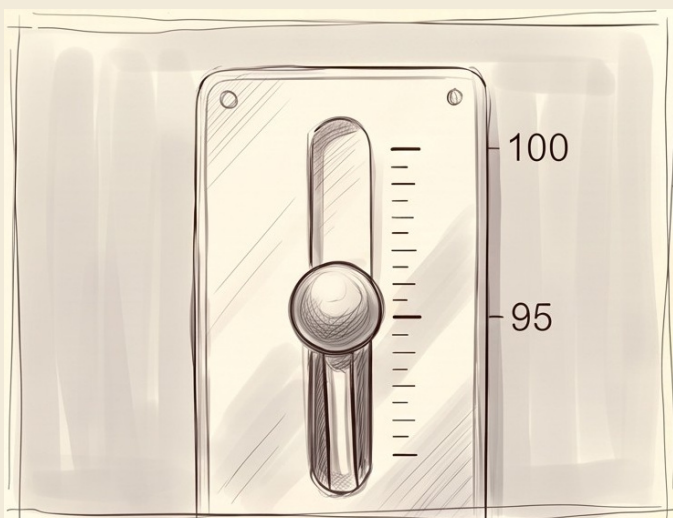
THE RELEASE



6.1 DRAWN, nobody needed on the day



6.2 DRAWN, nobody needed on the day



6.3 DRAWN, nobody needed on the day

The check comes back green

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

Coach Brain at the console running his safety check, every reading coming back green. He chooses. He is not overpowered and he is not tricked, and this is the single most important gesture in the film.

MANAN

COACH BRAIN: "All right. Everything's holding. Let's give him a little more."

His hand on the lever

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

The lever with his hand on it, and it must start LOW on the scale, well below 95. As boarded it already sits high, which makes the move across this panel and the next read as travelling downward and reverses the meaning of the whole scene. Low here, higher in the next frame.

MANAN

Nothing under this.

Eased open, and never to a hundred

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

The lever moves up and stops at 95. The gap of open track between the ball and 100 stays visible and is the moral architecture of the film. Protect it in every frame it appears.

MANAN

Caption, small: Never to one hundred.
Voice over. "More fibres recruited. Notice, still not all of them."



6.4 DRAWN, nobody needed on the day

Fibres lighting

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Muscle fibres lighting in scattered clusters inside a semi transparent leg. In sequence, never all at once, and never all of them.

MANAN

Voice over continues.



6.5 DRAWN, nobody needed on the day

More of them, and still not all

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

More clusters lit, and a clear proportion still dark. The reserve is inside the runner and it was always his. Nothing is given to him from outside.

MANAN

Nothing.



6.6 DRAWN, nobody needed on the day

Bodies finally permitted

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Traceur rooftop to rooftop, a dancer turning in the air, a swimmer leaving the wall. Each one higher in frame than the last. Rapid and rising. This is where the audience gets goosebumps.

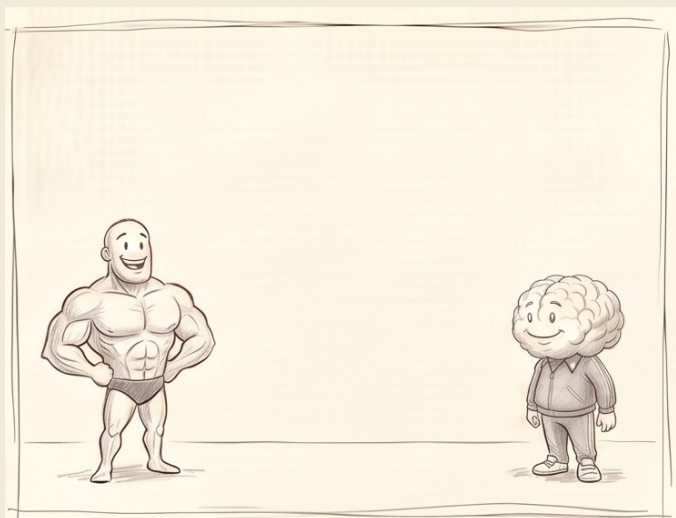
MANAN

Voice over. "The brain didn't make new energy. It gave permission."

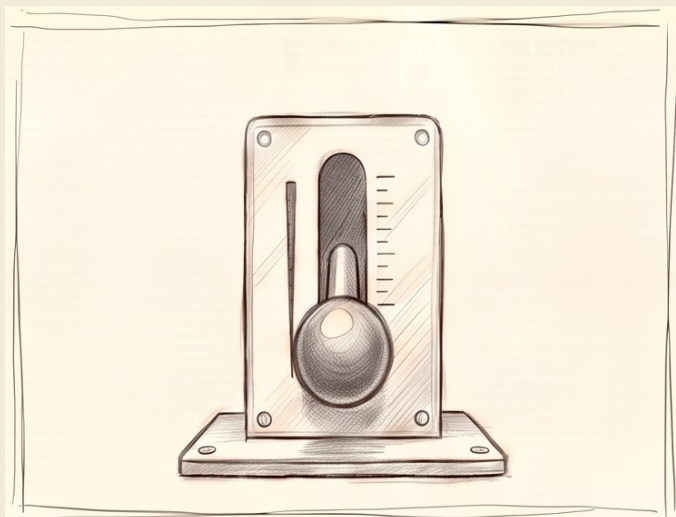
OPTIONAL AND WORTH FIVE MINUTES ON THE DAY:

SETUP D, Manan jumping on the spot against grey, arms out, laughing, high frame rate if the camera allows. It is not boarded, it costs almost nothing, and it gives the edit the option of putting him inside this moment.

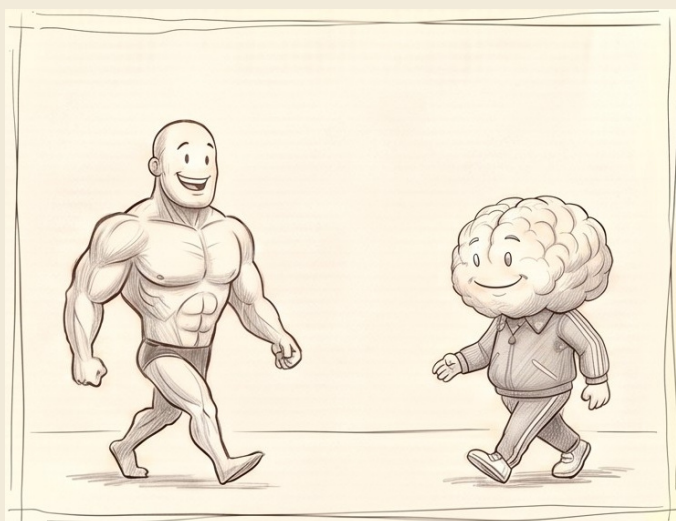
THE VERDICT



7.1 DRAWN, nobody needed on the day



7.2 DRAWN, nobody needed on the day



7.3 DRAWN, nobody needed on the day

Everything drains to white

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

The Muscle and Coach Brain a long way apart in a white field, facing each other. Stillness after the release. Do not rush this.

MANAN

Nothing. The silence is doing the work.

The brake between them

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

The lever alone in the white, standing between the two of them. The object they have been arguing about for a hundred years.

MANAN

Nothing.

They walk toward each other

CAMERA · VENKATESH

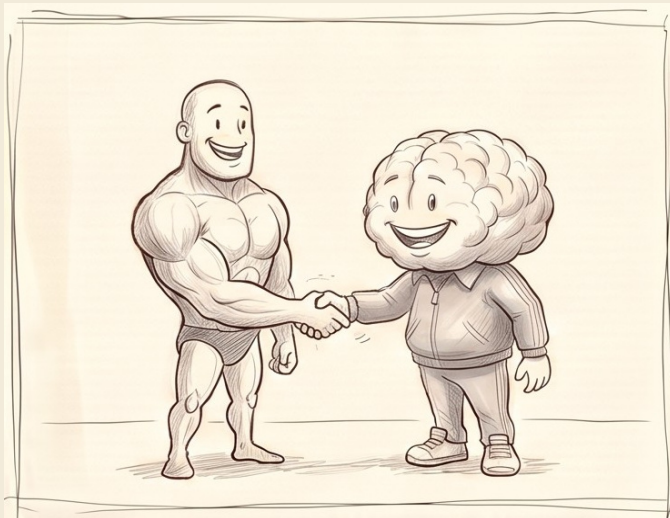
Nothing on the day. This frame is drawn.

ANIMATION · KRISTIJAN

Both walking in, easy and unhurried. No winner is being decided here.

MANAN

Voice over begins. "Hill was right about the muscle. Noakes was right about the brain."



7.4 DRAWN, nobody needed on the day

The handshake

CAMERA · VENKATESH

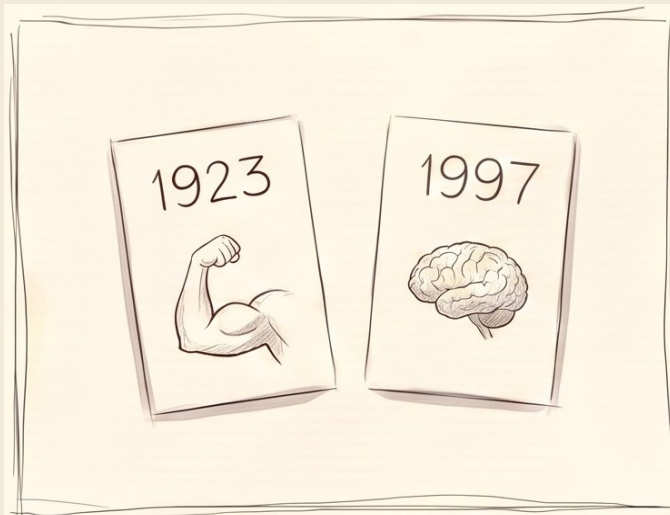
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

They shake hands. This is the answer to the question the film opened with, and it is given as a gesture rather than as a statement.

MANAN

Voice over continues.



7.5 DRAWN, nobody needed on the day

1923 and 1997

CAMERA · VENKATESH

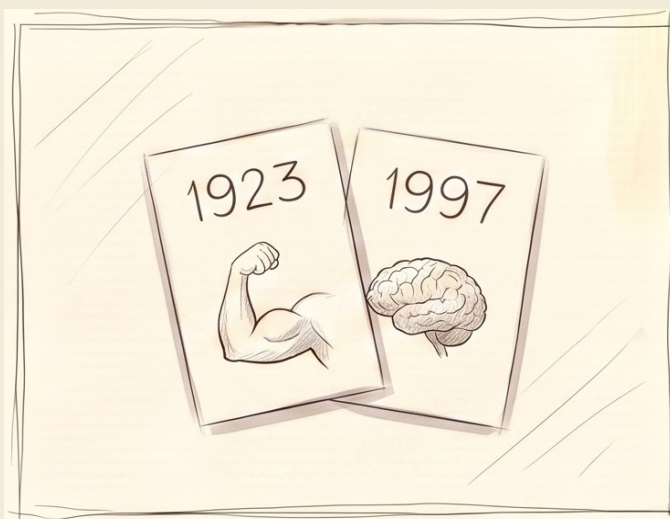
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

Two cards, an arm on one and a brain on the other, dated. Clean and plain.

MANAN

Voice over. "Today the evidence says both. And researchers are still working out exactly how."



7.6 DRAWN, nobody needed on the day

Still being tested

CAMERA · VENKATESH

Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

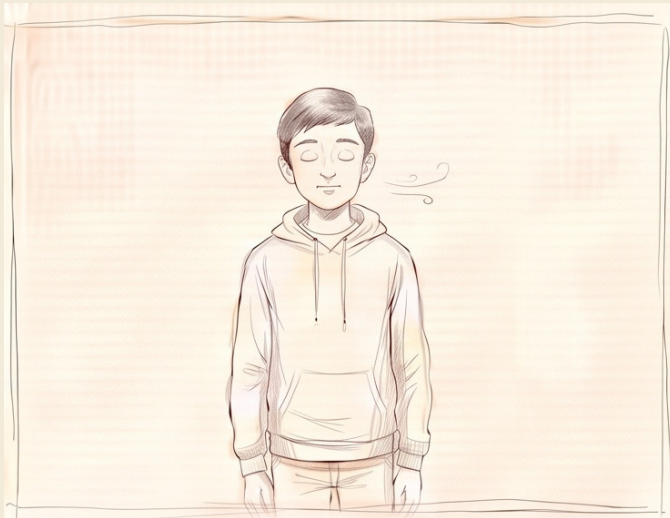
The two cards overlapping in the white. The most credible seconds in the film, because the film admits what is not yet settled.

MANAN

"Great ideas aren't accepted because they sound convincing. They're accepted because scientists keep testing them."

Manan's own line, kept word for word from his first draft.

THE INVITATION



8.1 LIVE ACTION + DRAWN

Alone, breathing

CAMERA · VENKATESH

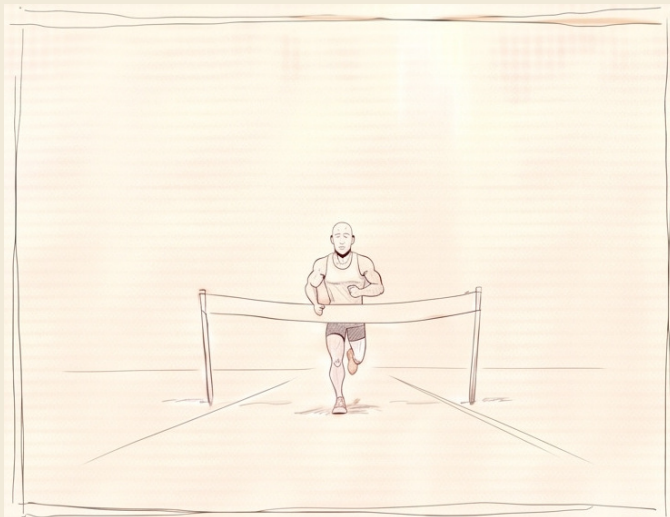
SETUP C. White void, flat even light from both sides, no modelling and no shadow on the backdrop. This is the brightest and the calmest setup of the day. Roll a full unbroken minute and use the stillest twenty frames.

ANIMATION · KRISTI JAN

White all round him. Nothing drawn in this frame at all except the world he is standing in.

MANAN

Arms at his sides, eyes closed, simply breathing. No performance of any kind. This is the hardest thing on the call sheet and the reason it is shot last, when he has stopped trying.



8.2 DRAWN, nobody needed on the day

The finish line

CAMERA · VENKATESH

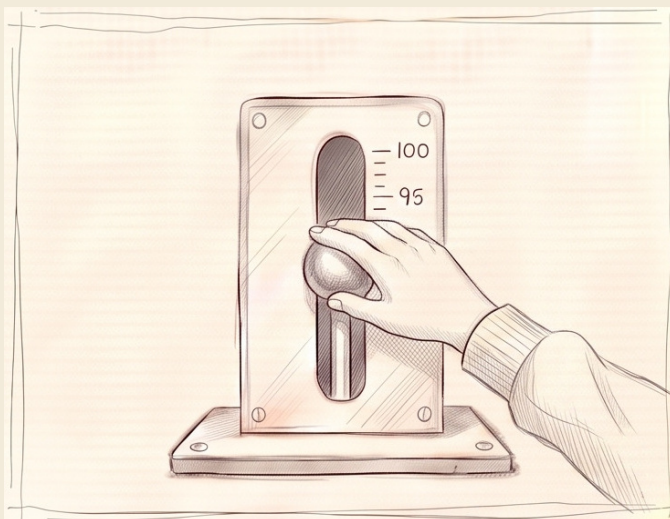
Nothing on the day. This frame is drawn.

ANIMATION · KRISTI JAN

The runner crossing, seen down the road from scene one. The film returns to where it started and the question has been answered.

MANAN

Nothing.



8.3 LIVE ACTION + DRAWN

His own hand on the lever

CAMERA · VENKATESH

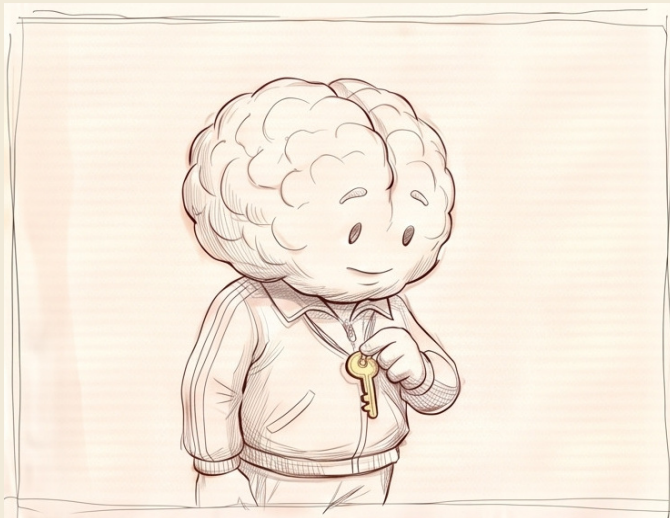
SETUP C, insert. His hand alone, resting on a real ball at chest height, so the fingers and the weight are right. A few seconds, held.

ANIMATION · KRISTI JAN

The lever drawn around his hand. It must be his hand and not an adult's, and the ball must sit at 95 with the short gap open above it. A real hand from the world of evidence reaching into the world of explanation is the thesis of the film and the climax of the story at the same time.

MANAN

Just the hand. Steady, unhurried, no grip and no drama.



8.4 DRAWN, nobody needed on the day

Coach Brain takes off the key

CAMERA • VENKATESH

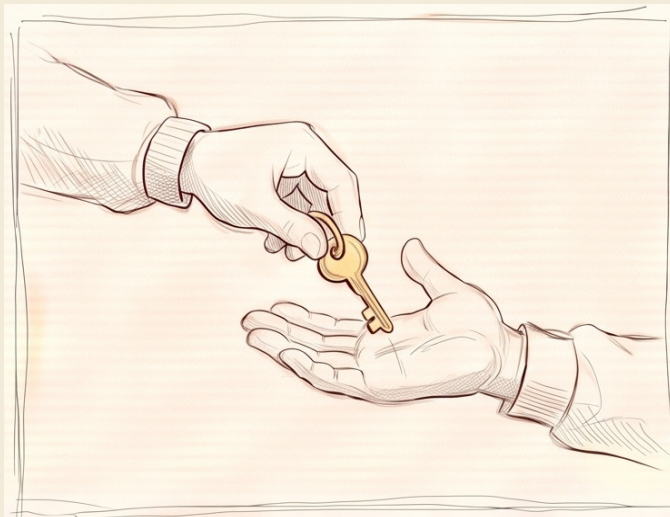
Nothing on the day. This frame is drawn.

ANIMATION • KRISTI JAN

He lifts the small gold key from his own neck. No fanfare, no ceremony. He is handing over a job.

MANAN

Nothing.



8.5 LIVE ACTION + DRAWN

The key handed across

CAMERA • VENKATESH

SETUP C. Receiving at chest height, then looking up. Give him a real key to hold so the hand knows what to do.

ANIMATION • KRISTI JAN

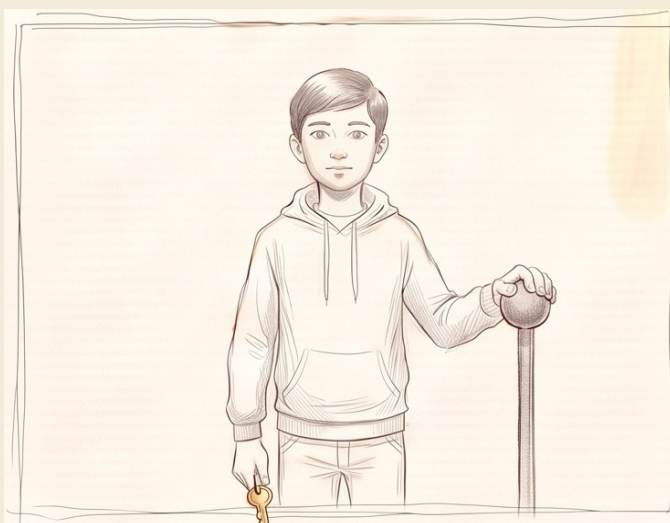
Coach Brain's drawn hand and Manan's photographed palm in one frame. The gold of the key is the only colour anywhere in the film.

MANAN

He takes it.

"The wall is real. But somebody set it."

Certain and unhurried, never triumphant. One sentence per take.



8.6 LIVE ACTION + DRAWN

Eyes open

CAMERA • VENKATESH

SETUP C. Wider, to lens, key in one hand and the other resting on the lever. Shoot the last line several times and let the last take be the tired one.

ANIMATION • KRISTI JAN

The lever beside him, the key in his hand. Then black, and the end card.

MANAN

"And what your brain believes is safe can be trained."

Then his own closing line, kept from his first draft:

"So the next time you watch an athlete find one last burst of speed, don't just wonder how strong their muscles are. Wonder what their brain believed was still safe."

Two things still open

Scene five has no frame with Manan in it. In the earlier breakdown he turned to camera at the end of that scene and said that changing what the brain believes opens the door. It is the most cheerful line in the film. Either it becomes voice over, or one frame of scene five turns to him. Worth a decision before the shoot rather than after, because it costs nothing on the day and cannot be added later.

Coach Brain has four spoken lines. He can be voiced by Manan pitched differently, or by somebody else entirely. Whoever it is, he is never sly and never a villain. He is a teacher who is pleased to have been found.

The wall is real. But somebody set it.